

Decision Tree Student Result Predictor ■

A simple Machine Learning project that uses a Decision Tree Classifier to predict whether a student will Pass or Fail based on study hours.

■ Project Overview

This project demonstrates a basic supervised machine learning model using Python and Scikit-learn. The model learns the relationship between a student's study hours and their result (Pass or Fail), then predicts the result for a new number of study hours entered by the user.

■ Objective

The main objective is to use a Decision Tree Classification algorithm to predict a student's academic result from the number of hours they study.

■ Dataset

Input feature: Student study hours

Target: Student result

The sample dataset contains study hours from 1 to 8 hours. Students studying fewer hours are labelled Fail, while students studying more hours are labelled Pass.

```
x = [[1],[2],[3],[4],[5],[6],[7],[8]]  
y = ["Fail","Fail","Fail","Fail","pass","pass","pass","pass"]
```

■■ Technologies & Libraries Used

- Python 3
- Scikit-learn
- DecisionTreeClassifier
- train_test_split
- accuracy_score

■ Machine Learning Algorithm

Decision Tree Classifier

A Decision Tree is a supervised machine learning algorithm used for classification. It learns decision rules from training data and uses those rules to predict the class of new data.

■ How the Program Works

1. The program creates a small dataset containing study hours and student results.
2. The data is divided into training and testing sets.
3. A Decision Tree Classifier is created.
4. The model is trained using the training data.
5. The user enters the number of hours a student can study.
6. The trained model predicts whether the student will Pass or Fail.
7. The model's accuracy is calculated using the test data.

■ Important Code

```
model = DecisionTreeClassifier()
```

```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.25, random_state=42
)
model.fit(x_train, y_train)

hours = int(input("How many hours student can do study? "))
prediction = model.predict([[hours]])
print("prediction:", prediction[0])
```

■■ How to Run

1. Install Python 3.
2. Install Scikit-learn using the terminal:

```
pip install scikit-learn
```

3. Save the Python program, for example as **project_2.py**.
4. Open the project in VS Code.
5. Run the program using:

```
python project_2.py
```

■ Example

```
How many hours student can do study? 6
prediction: pass
accuracy: 100.0 %
```

■ Accuracy

The program uses **accuracy_score()** to compare the model's predictions with the actual test results. Because this is a very small sample dataset, the reported accuracy should not be considered a reliable measure of real-world student performance.

■■ Note About the Code

In the provided screenshot, the **train_test_split()** line appears commented out. For the program to run correctly, that line must be active because **x_train, x_test, y_train, and y_test** are used later in the program.

■ Future Improvements

- Use a larger and more realistic dataset
- Add attendance, assignments, previous marks, and other features
- Improve data preprocessing
- Compare Decision Tree with other ML algorithms
- Create a graphical user interface
- Save and load the trained model

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